King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam I Term 222

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. Let $f(x) = \frac{1}{2}x\sin(2x)$, then $f'\left(\frac{\pi}{4}\right) =$

(a) $\frac{1}{2}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{4} + \frac{1}{4}$

(d) $\frac{1}{2} - \frac{\pi}{4}$

(e) $\frac{1}{2} + \frac{\pi}{4}$

2. If $f(x) = \begin{cases} \cos x, & x < 0 \\ ax + b, & x \geq 0 \end{cases}$ is differentiable everywhere, then $b - a =$

(a) 2

(b) -1

(c) 1

(d) 0

(e) -2

3. One statement is true about

$$g(x) = \frac{x^2 + x - 6}{|x - 2|}$$

(a) $\lim_{x\to 2^-} g(x) = -\infty$

(b) $\lim_{x\to 2} g(x)$ does not exist

(c) $x = 2$ is a vertical asymptote

(d) $\lim_{x\to 2^-} g(x) = \lim_{x\to 2^+} g(x)$

(e) $\lim_{x\to 2^+} g(x)$ does not exist

4. If the line $y = -4x + 1$ is tangent to the graph of the function $f(x) = k - x^2$, then $k =$

(a) $k = 4$

(b) $k = 3$

(c) $k = 2$

(d) $k = -3$

(e) $k = -4$

5. $\lim_{x\to 0} \frac{\sin x\,(1-\cos x)}{x^2} =$

(a) $-\infty$

(b) 0

(c) ∞

(d) 1

(e) -1

6. $\lim_{x\to \ln 3} \frac{e^{3x}-27}{e^{2x}-9} =$

(a) 1

(b) $\frac{9}{2}$

(c) 0

(d) 9

(e) $\ln 3$

7. If the constant a will make the function

$$f(x) = \begin{cases} ae^{x-1} + 3, & x < 1 \\ \tan^{-1}(x-1) + 2, & x \geq 1 \end{cases}$$

continuous over $(-\infty, \infty)$, then the value of $a^2 + a$ equals

(a) 0

(b) 1

(c) $\frac{-1}{e^2}$

(d) $\frac{1}{e^2}$

(e) -1

8. The domain of $f(x) = \dfrac{1}{\lfloor x \rfloor - x}$, where $\lfloor\ \rfloor$ denotes the greatest integer function, is

(a) All non-negative rational numbers

(b) All real numbers except zero

(c) Set of Natural numbers only

(d) All real numbers except the positive integers

(e) All real numbers except the integers

9. $\lim_{x\to 0} \frac{\cos x - 4\sin 3x - 1}{2x} =$

(a) -4

(b) 0

(c) 4

(d) 6

(e) -6

10. Let $f(x) = \dfrac{x \sin 2x}{\sin^2 x}$, which of the following is false?

(a) $\lim_{x\to 0} f(x) = 2$

(b) The graph of f has vertical asymptote of $x = \pi$

(c) The graph of f has vertical asymptote of $x = 0$

(d) f is discontinuous at $x = 0$

(e) The graph of f has vertical asymptote of $x = 2\pi$

11. If $\lim_{x\to 4} \left(4 - \frac{x}{2}\right) = 2$, then the value of δ such that $|f(x) - L| < \epsilon$ whenever $0 < |x - 4| < \delta$ is

(a) ϵ

(b) 2ϵ

(c) $\dfrac{\epsilon}{4}$

(d) 4ϵ

(e) $\dfrac{\epsilon}{2}$

12. The vertical asymptotes of the graph

$$f(x) = \frac{\ln(x^2 - 4)}{(x-1)(x+4)}$$

are

(a) $x = 1,\ x = -4$

(b) $x = 1$

(c) $x = \pm 2,\ x = 1$

(d) $x = \pm 2,\ x = -4$

(e) $x = \pm 2,\ x = -4,\ x = 1$

13. Let $f(x) = \sqrt{2x} - 4$, then $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} =$

(a) $\dfrac{1}{\sqrt{x}}$

(b) $\dfrac{2}{\sqrt{x}}$

(c) $\dfrac{1}{\sqrt{2x}}$

(d) $2\sqrt{x}$

(e) $\dfrac{1}{2\sqrt{x}}$

14. Let $p(x) = \dfrac{g(x)+1}{f(x)}$, $g(2) = -1$, $g'(2) = 2$, $f(2) = 3$, then $p'(2) =$

(a) $\dfrac{1}{3}$

(b) $\dfrac{4}{9}$

(c) $\dfrac{2}{3}$

(d) 0

(e) $\dfrac{5}{9}$

15. $\lim_{x\to\frac{1}{2}^-}\left(x^2\sec\pi x\right)=$

(a) ∞

(b) $\dfrac{1}{2}$

(c) $\dfrac{1}{4}$

(d) 0

(e) $-\infty$

16. $f(x)=\dfrac{\cos x}{1-\sin x}$, then $f'(x)=$

(a) 0

(b) $\dfrac{1}{1-\sin x}$

(c) $\dfrac{1}{(1-\sin x)^2}$

(d) $\dfrac{-\sin x}{1-\sin x}$

(e) $\dfrac{\sin x}{(1-\sin x)^2}$

17. $\lim_{x\to\pi/6}\dfrac{\dfrac{1+\csc x}{1-\csc x}+3}{x-\dfrac{\pi}{6}}=$

(a) $-\pi\sqrt{3}$

(b) $2\sqrt{3}$

(c) $-4\sqrt{3}$

(d) $\pi\sqrt{3}$

(e) $-2\sqrt{3}$

18. One statement is True about $f(x)=|\sin x|$ and $g(x)=\sin|x|$

(a) The domain of $g'(x)$ is $(-\infty,\infty)-\{0\}$

(b) The domain of $f(x)$ is $[0,\infty)$

(c) The range of $g(x)$ is $[0,1]$

(d) The range of $f(x)$ is $(0,1]$

(e) The domain of f' is all $x\neq n\pi$ where n is positive integer

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$f' = \frac{1}{2}\sin 2x + x\cos 2x$, so $f'(\pi/4) = \frac{1}{2}$.
2	**(c)**	Continuity gives $b = 1$, matching slopes gives $a = 0$, so $b - a = 1$.
3	**(b)**	$g \to 5$ from the right and $g \to -5$ from the left, so the two-sided limit does not exist.
4	**(d)**	$f' = -2x = -4 \Rightarrow x = 2$; $k - 4 = -7$, so $k = -3$.
5	**(b)**	$\frac{\sin x}{x} \to 1$ and $1 - \cos x \to 0$, so the limit is 0.
6	**(b)**	$u = e^x \to 3$: $\frac{u^2+3u+9}{u+3} \to \frac{27}{6} = \frac{9}{2}$.
7	**(a)**	$a + 3 = 2$ gives $a = -1$, so $a^2 + a = 0$.
8	**(e)**	$\lfloor x \rfloor - x = 0$ exactly at the integers, so exclude the integers.
9	**(e)**	$\frac{1}{2} \cdot 0 - 2 \cdot 3 = -6$.
10	**(c)**	$f = \frac{2x\cos x}{\sin x} \to 2$ at 0 (removable), so $x = 0$ is not an asymptote.
11	**(b)**	$\lvert f(x) - 2\rvert = \frac{1}{2}\lvert x - 4\rvert < \epsilon$ when $\lvert x - 4\rvert < 2\epsilon$, so $\delta = 2\epsilon$.
12	**(d)**	Domain $\lvert x\rvert > 2$: asymptotes at $x = \pm 2$ (log) and $x = -4$ (denominator); $x = 1$ is outside the domain.
13	**(c)**	Rationalize: $\frac{2}{\sqrt{2x+2h}+\sqrt{2x}} \to \frac{1}{\sqrt{2x}}$.
14	**(c)**	Quotient rule with $g(2) + 1 = 0$: $p'(2) = \frac{2\cdot 3}{9} = \frac{2}{3}$.
15	**(a)**	$\cos \pi x \to 0^+$ and $x^2 \to \frac{1}{4} > 0$, so the limit is $+\infty$.
16	**(b)**	Numerator $= -\sin x + \sin^2 x + \cos^2 x = 1 - \sin x$, so $f' = \frac{1}{1-\sin x}$.
17	**(c)**	Difference quotient of $F = \frac{1+\csc x}{1-\csc x}$ at $\pi/6$: $F' = -4\sqrt{3}$.
18	**(a)**	$g = \sin\lvert x\rvert$ has a corner at 0 only, so g' exists for all $x \neq 0$.